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# SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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## ABSTRACT

Fingerprinting Large Language Models (LLMs) underpins provenance verification and model attribution, yet the prevailing methodology rests on a demonstrably fragile premise: existing fingerprints are validated almost exclusively after fine-tuning, once a model has already accreted stable signatures from its training data, optimization dynamics, and hyperparameters. Because the overwhelming majority of a model’s capacity and knowledge is acquired during pretraining rather than downstream fine-tuning, large-scale pretraining—not fine-tuning—is the decisive regime in which lineage must be established. We demonstrate that existing fingerprinting methods are *unreliable* precisely in this regime: the post-hoc signatures they exploit materialize only after substantial training, so that their dependence on such signatures violates the classical Galtonian conception of a fingerprint as an identity intrinsic to, and persistent from, the moment of birth. We therefore introduce **SeedPrints**, a stronger and genuinely intrinsic notion of LLM fingerprinting that converts the prediction biases imprinted by random initialization into persistent, seed-dependent identifiers already present before a single optimization step is taken. We establish that untrained models exhibit reproducible, seed-determined prediction biases, and that these signals—individually weak yet statistically robust—remain detectable throughout training, yielding high-confidence lineage verification through a calibrated statistical test rather than a heuristic similarity threshold. Where prior techniques fail during early pretraining or collapse under distribution shift, **SeedPrints** remains effective across the entire training lifecycle, from initialization through large-scale pretraining to downstream adaptation. On LLaMA-style and Qwen-style models we establish seed-level distinguishability and birth-to-lifecycle identity verification; on large-scale pretraining trajectories and a real-world fingerprinting benchmark we further confirm robustness under prolonged training, pronounced domain shift, and the full range of parameter-altering deployment transformations.

Taken together, these results establish that initialization alone imprints a unique and durable identity on an LLM—a true “Galtonian” fingerprint—and that a single framework suffices to unify birth-stage and deployment-stage verification. Code will be released upon acceptance.

## 1 INTRODUCTION

LLM fingerprints have become a principled instrument for identifying, attributing, and tracing large language models through their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). Their purpose is to furnish model owners with a verifiable link between a suspicious model and its putative original, thereby enabling the detection of model theft and unauthorized reuse (Yoon et al., 2025; Zhang, 2025). As foundation models mature into high-value assets whose weights are routinely shared, adapted, and redistributed, the decisive question is no longer whether such a link can be recovered after deployment, but whether it holds across the *entire* arc of a model’s existence—from initialization, through large-scale pretraining, to downstream adaptation. This is the question our work answers in the affirmative.

## SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

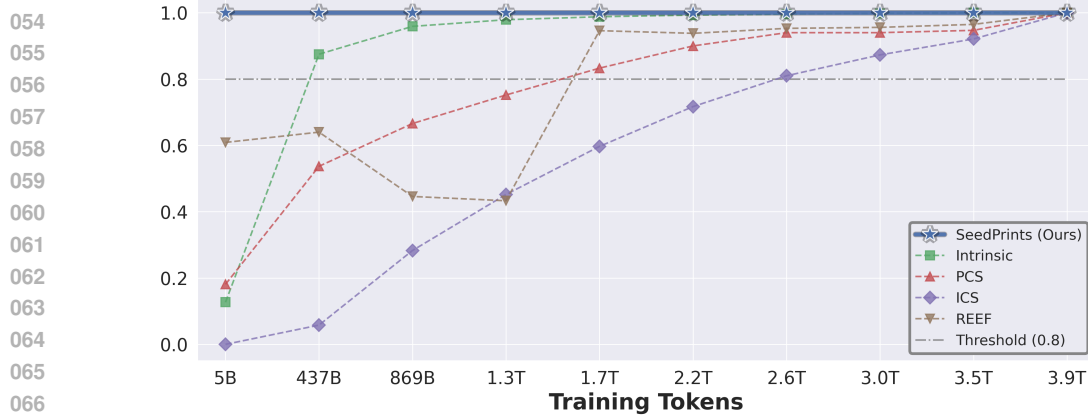


Figure 1: **Existing fingerprinting methods fail to detect model lineage during early pretraining, whereas SeedPrints detects it from the first checkpoint onward.** We compare five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each tested against the final checkpoint. The  $y$ -axis reports the similarity score (higher indicates a stronger lineage signal), and the dashed line marks the 0.8 detection threshold. Every baseline degrades and falls below the threshold at early checkpoints, exposing a systematic blind spot that pervades the pretraining regime; **SeedPrints**, by contrast, attains perfect detection ( $p \ll 0.001$ , plotted as  $1 - p$ ) from the very first checkpoint and remains saturated across the entire trajectory.

Much of this literature explicitly borrows the metaphor of biological fingerprints from Francis Galton’s *Finger Prints* (1892) (Galton, 1892), and we contend that the metaphor must be taken at its word:

*“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”*

The analogy imposes a precise and demanding standard: an effective fingerprint must be simultaneously unique and persistent, present from the very moment of a model’s “birth” at initialization. Judged against this standard, essentially all existing fingerprinting approaches fall short (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As Figure 1 makes concrete, these methods cannot reliably detect lineage during early pretraining. The failure is structural rather than incidental: because each method is defined only once a model has largely converged—reading patterns off settled parameters or generated text—the separability it achieves reflects not a birthmark of the model itself but an imprint of surface-level training components such as data signatures, optimization dynamics, and hyperparameters. This confound carries a direct empirical consequence: as we show in Section 5.1, existing baselines are readily misled and fail to identify true lineage once the training data distribution shifts substantially. Such methods therefore operate as *post hoc* identifiers rather than **Galtonian fingerprints**—the innate, immutable marks that accompany a model from its inception.

In this work, we advance a stricter and more principled notion of LLM fingerprinting: an intrinsic property fixed at *initialization* and detectable at *any* point during subsequent training. This reconception closes the pretraining blind spot that undermines prior work. We substantiate it through four contributions, which together trace a single argument from the diagnosis of the problem to a method that resolves it and its validation across the full model lifecycle:

- **Exposing a fundamental limitation of existing methods.** We establish that prior fingerprinting approaches cannot reliably detect lineage during early pretraining and are readily misled by substantial shifts in the training data distribution, demonstrating that their apparent success rests on training-induced artifacts rather than on intrinsic identity. (Section 5.1)
- **Discovering and operationalizing initialization-driven fingerprints.** We identify a previously overlooked phenomenon: untrained models already exhibit seed-dependent biases in their internal representations—a weak but persistent identity signal fixed at initialization (Section 3). Building directly on this finding, we introduce **SeedPrints**, a method that isolates these initialization-born fingerprints and, by construction, is invariant to confounding factors such as data distribution, training duration, and optimization dynamics. (Section 4)

- **Rigorous empirical validation.** We demonstrate that our method separates models differing *only* in initialization seed—even under identical training pipelines and data orders—and that this fingerprint persists under continual training across markedly diverse datasets, establishing that it tracks initialization rather than data. (Section 5.1)
- **Strong performance across realistic settings.** Our method detects lineage reliably throughout large-scale pretraining trajectories (e.g., OLMo-2-7B Stage 1) with no early-stage failure, and it remains decisive under large-scale fine-tuning (up to 700B tokens) and practical deployment on the *LeaFBench* benchmark (Shao et al., 2025), which comprises 65 models spanning 7 mainstream model families and 6 transformations (instruction tuning, fine-tuning, PEFT, quantization, model merging, and distillation). In these settings our method matches the strongest baseline (near-perfect) while substantially outperforming the remainder, thereby uniting birth-stage and deployment-stage verification within a single framework. (Sections 5.2 and 5.3)

## 2 RELATED WORK

Approaches to LLM protection fall into two complementary families: (i) *watermarking / active fingerprinting* methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* a signature from a model’s pre-existing behaviors without modifying it (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025). Our method belongs to the second family, yet it departs from every prior member along a defining axis that reorganizes the entire design space: the developmental stage at which the identifying signal first comes into existence.

**Watermarking and Active Fingerprinting.** Active approaches deliberately implant verifiable identifiers to enable subsequent ownership checks. Classic techniques include backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018) and digital signatures or hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, two forms predominate: *text watermarks*, which bias generation or insert predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.), and *model weight watermarks*, which embed identifiers into parameters or bind them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints are conceptually simple and can survive moderate fine-tuning (e.g., Dasgupta et al. 2024), these invasive schemes presuppose control of the training process and are therefore inapplicable to the retroactive marking of third-party models—precisely the setting in which provenance disputes actually arise.

**Passive LLM Fingerprinting.** Passive fingerprinting, by contrast, identifies models by analyzing their intrinsic properties without any modification, and the available techniques are naturally organized by the level of model access they assume. Under **white-box access**, signatures are extracted directly from model weights, exploiting properties such as the distribution of attention matrices (Yoon et al., 2025), the kernel alignment of internal representations (Zhang et al., 2024), or the stable direction of parameter vectors. In the **black-box setting**, fingerprinting instead analyzes input–output behavior through crafted queries (Pasquini et al., 2024), unique prompt–response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025), or iterative prompting–response games (Iourovitski et al., 2024), although such behavioral signals are typically less robust to fine-tuning. The decisive commonality is that all of these methods define their signature *post hoc*: each reads off emergent properties of a *completed* training process rather than the innate, “Galtonian” fingerprint—present from random initialization—that our work identifies and exploits. As a direct consequence, *none is designed for lineage verification during large-scale pretraining*: the very regime in which post-training signals have not yet formed is exactly the gap our method is built to close.

## 3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

In this section we establish the empirical foundation of our method: language models exhibit pronounced prediction biases that originate in the random initialization seed, and these biases remain statistically detectable even after training. This observation is the linchpin of the entire approach, and we develop its two constituent claims in turn—first that initialization induces a seed-specific bias, and then that training preserves it.

**Initialization induces seed-specific bias profiles.** We evaluate a LLaMA-2–style model initialized with seed 123 on 10,000 random input sequences of length 1,024, with tokens sampled uniformly

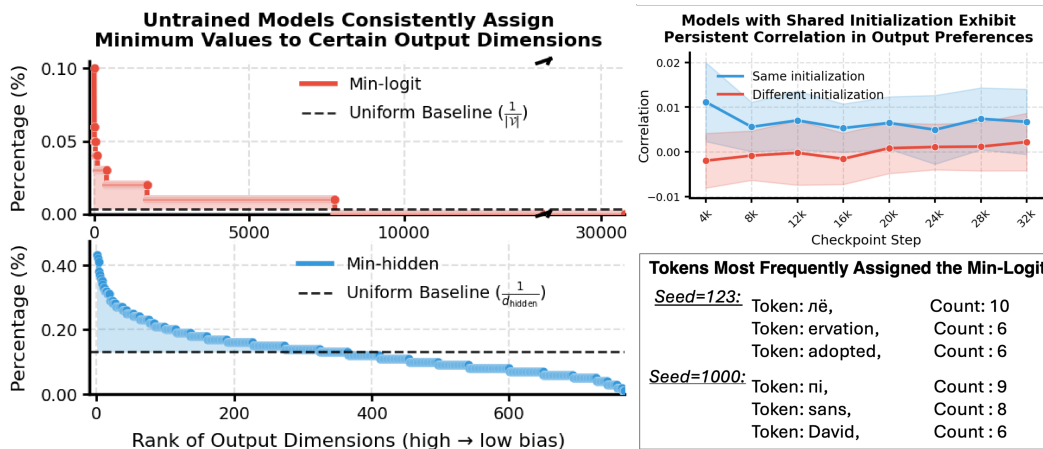


Figure 2: **Initialization-born output bias persists through training.** *Left:* Under fully random inputs, the outputs of a randomly initialized LLaMA-2-style model are far from uniform and instead exhibit a clear, systematic bias: certain dimensions are consistently disfavored, frequently attaining the minimum value across random inputs. This extreme bias is evident in both the logits (top, red) and the final hidden representations (bottom, blue). The dashed line marks the expected frequency under a uniform distribution, and the arrows in the top panel denote a broken  $x$ -axis that omits low-frequency tail ranks. *Upper Right:* Throughout training, a model remains weakly but reliably correlated in its output bias with the base model it was trained from. This correlation distribution is consistently shifted upward relative to correlations computed against independently initialized base models, establishing that initialization-born bias leaves a measurable signal that survives training. *Lower Right:* The most strongly biased dimensions are seed-determined, illustrated here through the tokens that most frequently receive the minimum logit.

from the vocabulary. In Figure 2 (*Left*), we plot the frequency with which each output token (upper, from the final logits) and each hidden dimension (lower, from the last-layer representations) attains the minimum value across the 10,000 random trials. The pronounced departure from uniformity in both panels is not incidental fluctuation but an extreme, systematic output-preference bias. Repeating the experiment across initialization seeds isolates the origin of this structure: while the overall *magnitude* of the bias is stable across seeds, the specific argmin dimensions are seed-determined (Figure 2, *Lower Right*)—establishing that the bias is a property of the seed, not of the architecture alone. This behavior is precisely what recent theory predicts: biases in randomly initialized models arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and further amplified by self-attention (Li et al., 2026). Under this account, distinct initialization seeds induce distinct contraction directions, which in turn produce the distinct argmin patterns we observe across output dimensions; this supplies a mechanistic explanation for why the fingerprint is inherently seed-specific.

**Training preserves initialization-born bias.** Although training substantially rescales output magnitudes, it does not erase the relative preference over output dimensions imprinted at initialization. We train the previously initialized model (seed 123) for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate its intermediate checkpoints. Restricting attention to the  $m = 50$  output dimensions most frequently assigned the minimum value by the initialized model, we compute, for each checkpoint, the correlation between its responses and those of the initialized model across random inputs on exactly these dimensions; this quantity measures how similarly the checkpoint and its initialization rank—and disfavor—these coordinates across inputs. As a control, we compute the identical correlation against an independently initialized model (seed 1000). As shown in Figure 2 (*Upper Right*), although the absolute correlations are small, the correlations for models sharing an initialization are consistently and systematically shifted upward relative to the independent control, and they stabilize rather than decay toward zero as training proceeds. Together, these results establish that initialization leaves a weak but statistically detectable bias signal that persists throughout training—exactly the property a lifelong fingerprint requires.

## 4 ALGORITHM

Section 3 establishes that models sharing an initialization lineage retain weak but consistent agreement on a specific subset of output dimensions—those most disfavored by the untrained model under random inputs. This observation translates directly into a precise design objective: given any two models, and *without* access to the true initialized model, recover the set of dimensions that carry this bias signal and test whether the two models’ agreement on them is statistically significant. Algorithm 1 summarizes the complete procedure; we specify each of its three components—dimension localization, correlation estimation, and hypothesis testing—in turn.

**Extracting identity dimensions between any two models.** Let  $X = \{x_i\}_{i=1}^n$ , where each  $x_i \in \mathbb{R}^{\ell \times d}$  denotes a random sequence of length  $\ell$ . Each random input  $x_i$  admits two equivalent instantiations—uniformly sampling  $\ell$  token IDs from the vocabulary, or directly sampling  $\ell$  independent vectors from a  $d$ -dimensional isotropic Gaussian distribution—and the two are interchangeable for our purposes. For any model  $g$ , we define its mean response vector as  $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$ , where  $d_{\text{out}}$  denotes the output dimensionality and  $g(x_i)$  denotes either the model’s logits or its final-layer representations. We then define its disfavored dimensions as the bottom- $m$  coordinates of  $\bar{g}$ , namely the size- $m$  index set that minimizes the aggregate mean response:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We adopt the mean response vector as the bias estimator because it is provably more stable and noise-robust than per-sample argmin frequency (i.e., counting how often each dimension attains the minimum value, as illustrated in Figure 2), while performing at least as well empirically; Appendix C.1 provides the formal comparison.

Given two models  $f$  and  $f'$ , we define their *identity dimensions* as the intersection  $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$ . The rationale is direct: two models sharing an initialization lineage should independently recover similar disfavored dimensions, and thus exhibit non-trivial overlap in their bottom- $m$  sets. The intersection criterion therefore functions as a form of mutual verification—it suppresses spurious correlations arising from architectural similarity or shared training dynamics and isolates the bias signal most plausibly attributable to shared initialization. When one model lies closer to initialization, its disfavored dimensions may equivalently serve as a direct anchor.

**Correlation statistics.** Restricting both models to the identity dimensions  $\mathcal{S}$ , we compare their responses across random inputs.<sup>1</sup> For each  $j \in \mathcal{S}$ , we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This yields a set of correlation statistics  $T = \{\tau_j\}_{j \in \mathcal{S}}$  that characterizes the full distribution of agreement between the two models across identity dimensions. Consistent with Section 3, same-lineage models exhibit distributionally higher agreement on these dimensions than unrelated models do.

**Hypothesis testing via a null distribution.** Because the correlation distributions of same-lineage and different-lineage models can overlap, thresholding an individual statistic is inadequate; we instead test whether the observed correlations are significantly larger than expected under the null hypothesis of no lineage relation. Ideally, this null would be estimated from the correlation distribution between independent models, as in Figure 2. Estimating such a baseline empirically, however, requires comparisons across many independently initialized models and is therefore computationally expensive. Appendix B shows that the null correlation distribution between independently initialized models is well approximated by a Gaussian, which motivates a substantially cheaper alternative: an empirical null constructed by passing Gaussian surrogates through the identical pipeline, as summarized in Algorithm 1.

This simulation-based null already furnishes a strong empirical approximation, yet it retains avoidable randomness and computational overhead: its precision is bounded by the number of simulation trials, its estimates fluctuate across runs, and comparing two empirical distributions typically imports

<sup>1</sup>Prior to computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This normalization eliminates scale differences across models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

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**Algorithm 1** Distribution Correlation Test on Identity Dimensions

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**Require:** base model  $f$ , suspicious model  $f'$ ; random inputs  $X = \{x_i\}_{i=1}^n$ ; fingerprint size  $m$ ; significance level  $\alpha$

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs  $\bar{f}, \bar{f}'$  over  $X$
- 2:  $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$  bottom- $m$  dimensions of  $\bar{f}, \bar{f}'$
- 3:  $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$  (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize  $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$  for each  $x_i \in X$  across identity dimension
- 5: **for** each  $s_j \in \mathcal{S}$  **do**
- 6:    $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$  (correlation in in input-wise preferences)
- 7: **end for**
- 8:  $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{S}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct  $\mathcal{T}_{\text{null}}$  by applying the same pipeline to two Gaussian matrices  $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test  $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$  vs.  $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage  $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

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additional assumptions from the chosen test (e.g., the one-sided  $t$ -test or the Mann–Whitney  $U$  test). Because the null distribution of Kendall–Tau under weak independence is available in closed form, we take a further step and replace the simulation with an exact analytical test, while shipping both the empirical and the analytical implementations in our code repository. Concretely, under the null each Kendall–Tau correlation is approximately zero-mean with known variance  $\sigma^2$ . By the central limit theorem, as the number of identity dimensions  $|\mathcal{S}|$  grows, the distribution of  $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$  is well

approximated by  $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$ . Although softmax normalization introduces weak cross-dimension dependencies, the relatively high temperature ( $T = 10$ ) attenuates them, so that the independence assumption remains an accurate approximation in practice; we confirm empirically that this Gaussian approximation closely matches the distribution obtained from the full pipeline (Appendix B). We accordingly compute the one-sided  $z$ -score  $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$  and the corresponding  $p$ -value  $p = 1 - \Phi(z)$ ,

where  $\Phi$  is the standard normal cumulative distribution function, and we reject the null—declaring shared lineage—whenever  $p < 0.01$ . The outcome is thus a calibrated statistical decision rather than a heuristic similarity comparison.

**Practical instantiation.** The model output may be taken as either logits or hidden representations, both of which are consistent with our formulation. Estimating bias patterns from logits, however, requires substantially more random inputs as the vocabulary grows, and thus incurs considerable computational overhead (Appendix C.2 provides the supporting theoretical analysis). For efficiency and stability, we therefore adopt final-layer hidden states as the default output. The resulting  $p$ -value constitutes a direct statistical test for lineage, dispensing entirely with heuristic similarity thresholds.

## 5 EXPERIMENTS

Our experiments are organized around two complementary claims. In Section 5.1 we establish that our method functions as a *genuine* fingerprint by demonstrating both properties the Galtonian standard demands: (i) *birth verification at the seed level* and (ii) *verifiability across the entire training lifecycle*. Existing baselines satisfy neither property: they cannot support verification at early pretraining stages, and they rely predominantly on data-dependent signals that collapse under substantial distribution shift. Unless stated otherwise, we report results using the analytical null with *hidden state* outputs; complementary results using the empirical null with both *logit* and *hidden state* outputs, evaluated under the one-sided  $t$ -test and the Mann–Whitney  $U$  test, appear in Appendix D.2 and reproduce every conclusion of the main paper without exception. In Section 5.3 we then evaluate all methods under practical infringement scenarios using *LeaFBench* (Shao et al., 2025): across 65 distinct model instances spanning 6 representative post-development techniques, our method matches the strongest baselines in post-training settings while remaining robust to the full range of deployment transformations—and it does so while additionally providing the birth-stage verification that those baselines cannot.

**Baselines.** We principally compare against four passive fingerprinting baselines, spanning both weight- and representation-based signals. *Intrinsic fingerprint* (Yoon et al., 2025) (denoted *PDF* in some papers) compares models through the similarity of the layerwise standard-deviation profiles of

Table 1: Fingerprint behaviors between models initialized with different seeds: every  $p$ -value exceeds 0.01, so SeedPrints never conflates models originating from distinct seeds.

| Model Pairs               | $p$ -value ( $> 0.01$ ) |
|---------------------------|-------------------------|
| $s_{42}$ VS. $s_{2000}$   | 0.116                   |
| $s_{123}$ VS. $s_{42}$    | 1.000                   |
| $s_{1000}$ VS. $s_{123}$  | 0.327                   |
| $s_{2000}$ VS. $s_{1000}$ | 1.000                   |

Table 2: Trained models retain the fingerprint of their initialization: SeedPrints yields  $p < 0.01$  for every  $s^{init}_{-}s^{base}$  pair, whereas all four baselines fail to detect the shared lineage in every single case.

| Model Pairs                             | SeedPrints | Baselines |        |        |        |
|-----------------------------------------|------------|-----------|--------|--------|--------|
|                                         | $p$ -value | Intrinsic | REEF   | PCS    | ICS    |
| $s^{init}_{42}$ vs. $s^{base}_{42}$     | 3.37e-26✓  | -0.021×   | 0.375× | 0.580× | 0.196× |
| $s^{init}_{123}$ vs. $s^{base}_{123}$   | 3.16e-33✓  | 0.149×    | 0.369× | 0.581× | 0.188× |
| $s^{init}_{1000}$ vs. $s^{base}_{1000}$ | 1.12e-21✓  | -0.252×   | 0.381× | 0.581× | 0.188× |
| $s^{init}_{2000}$ vs. $s^{base}_{2000}$ | 3.70e-31✓  | -0.337×   | 0.331× | 0.581× | 0.188× |

their attention parameters. **REEF** (Zhang et al., 2024) computes centered-kernel-alignment (CKA) similarity between the feature representations elicited by identical samples across two models. **PCS** and **ICS** (Zeng et al., 2024) (collectively *HuRef* in some papers) are weight-similarity methods: PCS flattens all parameters and measures cosine similarity, whereas ICS first constructs weight invariants and then measures cosine similarity on those invariants. In Section 5.3 we additionally include a weaker gradient-based fingerprint, **Gradient** (Shao et al., 2025). Following Zhang et al. (2024), every similarity-based baseline uses a **0.8 similarity threshold** for its binary decision.

For readability, cell colors in every experiment table encode ground-truth lineage: **green** marks models that share a source, whereas **red** marks models drawn from distinct sources. For instance,  $s^{init}_{42}$  vs.  $s^{base}_{42}$  compares a seed-42 initialization against its continued-pretrained descendant, and is therefore green;  $s^{init}_{2000}$  vs.  $s^{base}_{42}$  compares a seed-2000 initialization against a model trained from a seed-42 initialization, and is therefore red. A ✓ marks a correct detection and a × marks an error.

## 5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main paper reports LLaMA-style results and defers Qwen-style results to Appendix D; because the conclusions hold across both architectures, our findings are not artifacts of a particular model design.

**Different initialization seeds produce distinct fingerprints.** Table 1 reports  $p$ -values from our correlation test between pairs of models initialized with different random seeds (42, 123, 1000, and 2000). Every  $p$ -value exceeds 0.01, so our method never conflates models originating from different seeds: distinct seeds yield distinct fingerprint behaviors, and the models are cleanly separable “at birth.”

**Training preserves the initialization fingerprint.** Table 2 compares each initialization model  $s^{init}$  with its descendant  $s^{base}$  trained on OpenWebText (Gokaslan et al., 2019) ( $\approx 10$ B tokens). Across every seed-model pair the  $p$ -values are uniformly  $< 0.01$ : the bias profiles remain strongly correlated, and descendant and initialization are certified as sharing a common lineage—so the trained model demonstrably inherits the fingerprint of its initialization. The four baselines exhibit exactly the opposite behavior: without exception they fail to detect this shared lineage, which confirms that their separability derives from training-induced artifacts rather than from initialization itself.

**Identical data and order do not force fingerprints to converge.** In Table 3, all four “suspicious” models  $s^{base}_i$  for  $i \in \{42, 123, 100, 2000\}$  are trained on *exactly the same corpus* (OpenWebText) and in the *same data order*—we fix the training seed to lock the curriculum—so that their sole point of difference is the initialization seed  $i$ . Across every cross-seed pair the  $p$ -values remain  $> 0.01$ , in sharp contrast to the near-zero values of Table 2. This experiment isolates the seed as the causal source of the fingerprint: the signal remains seed-specific even when data and curriculum are held perfectly identical.

**Continual training on diverse datasets does not confound the fingerprint.** The experiments above establish that SeedPrints behaves as a biometric fingerprint. From a copyright standpoint, the inability of prior methods to separate models by seed is not in itself a shortcoming, since initialization seeds carry no clear copyright status. Their genuine fragility lies elsewhere: their attribution is easily misled

Table 3: Identical dataset and training order do not force fingerprint behaviors to coincide across different initializations: every  $p$ -value remains  $> 0.01$ , which isolates the initialization seed as the sole source of the fingerprint.

| Model Pair                              | $p$ -value ( $> 0.01$ ) |
|-----------------------------------------|-------------------------|
| $s_{42}^{init}$ vs. $s_{123}^{base}$    | 0.573                   |
| $s_{123}^{init}$ vs. $s_{2000}^{base}$  | 0.724                   |
| $s_{2000}^{init}$ vs. $s_{1000}^{base}$ | 0.109                   |
| $s_{1000}^{init}$ vs. $s_{42}^{base}$   | 0.883                   |

Table 4: Fingerprint persistence under continual training on diverse datasets (base model: seed 1000, corpus openwebtext). SeedPrints attributes lineage correctly under both TinyStories and the sharply shifted code corpus (The Stack), whereas every baseline breaks down on The Stack, tracking domain rather than lineage.

| Setting                 | SeedPrints    | Baselines |        |        |        |
|-------------------------|---------------|-----------|--------|--------|--------|
| Continual corpus (seed) | $p$ -value    | Intrinsic | REEF   | PCS    | ICS    |
| TinyStories (1000)      | $\approx 0$ ✓ | 1.000✓    | 0.759× | 0.999✓ | 0.996✓ |
| TinyStories (123)       | 1.000✓        | 0.950×    | 0.658✓ | 0.332✓ | 0.012✓ |
| the_stack (1000)        | $\approx 0$ ✓ | 0.489×    | 0.557× | 0.585× | 0.123× |
| the_stack (123)         | 1.000✓        | 0.445✓    | 0.580✓ | 0.301✓ | 0.026✓ |

by data distribution, *failing to recognize lineage when the training distribution shifts substantially during continued training.*

To isolate this failure mode, in Table 4 we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two deliberately dissimilar corpora: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We contrast (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (seed 123). Because the baselines are insensitive to initialization differences, the distractor base is pretrained with a different data order on OpenWebText and then continually trained on the *same* target corpus, so that the decisive question becomes whether an attribution method can identify which descendant genuinely shares lineage with the base.

The result is unambiguous: every prior baseline fails in the code setting (The Stack), misclassifying true descendants as distractors. This failure reveals that the baselines largely track *domain* similarity rather than lineage identity—TinyStories remains close in distribution to the pretraining corpus (OpenWebText), whereas The Stack diverges sharply, and such a pronounced shift alone suffices to defeat detection. Our method, by contrast, attributes lineage correctly under both corpora. SeedPrints is therefore not a proxy for data distribution: it withstands substantial domain shift and persists well beyond the initial pretraining stage.

## 5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

**Controlled pretraining trajectory.** Can our method reliably recognize its own offspring along an entire training trajectory? A controlled pretraining experiment on OpenWebText, in which the initialization seed is known, answers this affirmatively: testing lineage at intermediate checkpoints yields consistent detection at every stage (Appendix D.2.1, Figure 5).

**Large-scale pretraining.** To assess lineage under realistic foundation-model training, we analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B  $\rightarrow$  3.9T tokens), treating the final checkpoint as the target model and every earlier checkpoint as a tested ancestor. Unlike the controlled OpenWebText setting, this run spans heterogeneous corpora, far longer optimization, and large-scale representation drift—and therefore constitutes a stringent stress test of persistence. We plot  $1 - p$  for directional consistency with the similarity-based baselines; all of our  $p$ -values are  $\ll 0.01$  and numerically indistinguishable from zero.

Figure 1 delivers an unambiguous verdict: **every existing baseline fails to verify lineage during early pretraining** (e.g.,  $ICS \approx 0$ ,  $PCS < 0.3$ , and  $REEF < 0.6$  within the first **trillion** tokens), whereas SeedPrints remains verifiable from the very outset and only strengthens as training proceeds. This contrast exposes a critical and previously overlooked gap in existing evaluations: **prior work verifies lineage almost exclusively after fine-tuning, where detection is substantially easier.** Our results demonstrate that large-scale pretraining itself can dramatically reshape representations and *inflate* late-stage lineage signals, thereby conferring a *false sense of safety*. We therefore argue that lineage verification must prioritize the early stages of pretraining—when misuse is hardest to detect—rather than relying on late-stage checkpoints alone.

**Large-scale finetuning.** We further benchmark our method against the baselines under standard fine-tuning evaluations, testing suspect models fine-tuned from Llama-2-7b (the base model) at data

Table 5: Fingerprinting results under large-scale finetuning. Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the  $p$ -value from our correlation test ( $< 0.01$  indicates a strong lineage signal), whereas the four baselines report similarity scores (threshold = 0.8, higher is better). SeedPrints sustains decisive detection ( $p < 0.01$ ) across the entire 5M-to-700B-token range, including the 500B- and 700B-token regimes in which the weight-based baselines PCS and ICS fail outright.

| Model                               | # Tokens | SeedPrints $p$   | Intrinsic $\uparrow$ | REEF ( $\uparrow$ ) | PCS ( $\uparrow$ ) | ICS ( $\uparrow$ ) |
|-------------------------------------|----------|------------------|----------------------|---------------------|--------------------|--------------------|
| Llama-2-finance-7B (Heenan, 2023)   | 5M       | $10^{-41955}$ ✓  | 1.0000 ✓             | 0.9950 ✓            | 0.9979 ✓           | 0.9952 ✓           |
| Vicuna-1.5-7B (Chiang et al., 2023) | 370M     | $10^{-103043}$ ✓ | 1.0000 ✓             | 0.9985 ✓            | 0.9985 ✓           | 0.9949 ✓           |
| Wizardmath-7B (Luo et al., 2023)    | 1.8B     | $10^{-180550}$ ✓ | 1.0000 ✓             | 0.9979 ✓            | 1.0000 ✓           | 0.9994 ✓           |
| Meditron-7B (Chen et al., 2023)     | 48B      | $10^{-42566}$ ✓  | 0.9990 ✓             | 0.9978 ✓            | 1.0000 ✓           | 0.9817 ✓           |
| CodeLlama-7B (Meta AI, 2024)        | 500B     | $10^{-3552}$ ✓   | 0.9480 ✓             | 0.9947 ✓            | 0.6863 ✗           | 0.3369 ✗           |
| Llemma-7B (Azerbayev et al., 2023)  | 700B     | $10^{-5136}$ ✓   | 0.9470 ✗             | 0.9984 ✓            | 0.6682 ✗           | 0.2905 ✗           |

volumes ranging from 5 million to 700 billion tokens. The suspects span a deliberately diverse set of downstream variants—Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), Code-Llama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023)—at fine-tuning volumes of 5M, 370M, 1.8B, 13B, 500B, and 700B tokens, respectively. As Table 5 shows, our method maintains  $p < 0.01$  in every setting, sustaining decisive lineage detection even at fine-tuning budgets as large as 700B tokens—precisely the regime in which two of the weight-based baselines break down entirely.

### 5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployments routinely subject foundation models to *parameter-altering* adaptations—fine-tuning, PEFT, quantization, and beyond—that can weaken or distort fingerprint signals, and our method is engineered to remain effective precisely in this regime. To evaluate its reliability under these realistic conditions, we adopt *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing.

**Across Source Models.** We consider two categories of source model for auditing: *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each category, the task is to separate *derivative* models (produced by post-training adaptation) from *independent* models (trained without the source weights), mirroring the common auditing scenario in which a lineage claim must be tested against either a PT base or its IT variant. The evaluation spans 65 models in total; Appendix E enumerates every model name and HuggingFace repository.

**Metrics.** We quantify detection performance with (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), the latter measuring the maximum separation between the score distributions of derivative and independent models.

Our method emits per-model  $p$ -values from statistical tests, which are objects of a fundamentally different kind from the similarity scores used by the baselines: similarity scores are linear quantities whose differences carry proportional meaning, whereas  $p$ -values are tail probabilities that encode the rarity of an observation under the null and cannot be read on a linear scale. To enable comparison with baselines that compute AUC from similarity scores, we convert our  $p$ -values via  $s = 1 - p$ . If anything, this conversion handicaps our method—small  $p$ -values vary on an exponential scale while threshold sweeping is linear, which blurs fine-grained differences—yet even under this handicap our approach attains comparable or superior performance. We therefore (i) compute AUC for all methods by sweeping thresholds over the scores  $s$  and (ii) report the KS statistic as a threshold-free measure of distributional separability. This protocol preserves compatibility with the baseline pipelines while retaining the statistical interpretability of our method. We omit Manhattan distance, since absolute distances between  $p$ -values are not meaningful.

We emphasize that calibrating an optimal similarity threshold is infeasible in most practical settings. We report AUC purely for comparability, but the essential distinction remains decisive: our method delivers a direct statistical test. **For any pair of models, it outputs a  $p$ -value that supports a definitive, threshold-free decision on whether they share lineage**—precisely the property that enables reliable verification in deployment.

Table 6: Performance comparison of LLM fingerprinting methods across source models. “PT” and “IT” denote pre-trained and instruction-tuned source models, respectively. The per-family breakdown covers the 6 families possessing both PT and IT models. The “Overall” scores are computed over all test model pairs under a single global threshold, and are therefore not the average of the per-family scores. SeedPrints reaches near-saturated overall AUC/KS (0.992/0.986), matching the strongest deployment-oriented baselines (Intrinsic, ICS) and standing far above REEF and Gradient—while uniquely retaining birth-stage verifiability.

| Method     | Metric         | Qwen2.5-7B |       | Qwen2.5-14B |       | Llama3.1-8B |       | Mistral-7B |       | Gemma2-2B |       | Llama2-7B |       | Overall      |
|------------|----------------|------------|-------|-------------|-------|-------------|-------|------------|-------|-----------|-------|-----------|-------|--------------|
|            |                | PT         | IT    | PT          | IT    | PT          | IT    | PT         | IT    | PT        | IT    | PT        | IT    |              |
| REEF       | AUC ↑          | 0.796      | 0.862 | 0.947       | 0.913 | 1.000       | 0.999 | 0.997      | 0.997 | 0.963     | 0.968 | 0.750     | 0.706 | <b>0.915</b> |
|            | KS Statistic ↑ | 0.492      | 0.633 | 0.875       | 0.735 | 0.987       | 0.987 | 0.975      | 0.980 | 0.950     | 0.837 | 0.750     | 0.694 | <b>0.739</b> |
| Gradient   | AUC ↑          | 0.657      | 0.743 | 0.763       | 0.789 | 0.897       | 0.904 | 0.872      | 0.682 | 1.000     | 1.000 | 0.650     | 0.768 | <b>0.801</b> |
|            | KS Statistic ↑ | 0.354      | 0.423 | 0.500       | 0.563 | 0.646       | 0.705 | 0.725      | 0.500 | 1.000     | 1.000 | 0.275     | 0.479 | <b>0.508</b> |
| ICS        | AUC ↑          | 1.000      | 1.000 | 0.997       | 0.997 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.994</b> |
|            | KS Statistic ↑ | 1.000      | 1.000 | 0.975       | 0.979 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.943</b> |
| Intrinsic  | AUC ↑          | 1.000      | 1.000 | 1.000       | 1.000 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.994</b> |
|            | KS Statistic ↑ | 1.000      | 1.000 | 1.000       | 1.000 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.989</b> |
| SeedPrints | AUC ↑          | 0.992      | 0.994 | 0.988       | 0.990 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.992</b> |
|            | KS Statistic ↑ | 0.985      | 0.987 | 0.975       | 0.980 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.986</b> |

**Parameter-Altering Techniques.** To probe fingerprinting robustness across the full transformation landscape, we evaluate models produced via the six representative techniques that dominate contemporary deployment: (1) instruction tuning (Instruct), (2) general-purpose fine-tuning (Finetune), (3) parameter-efficient fine-tuning (PEFT), (4) quantization, (5) model merging (Merge), and (6) distillation.

### 5.3.1 FAMILY-WISE EFFECTIVENESS

The benchmark comprises seven model families. For each family we treat both its pretrained base model (PT) and its instruction-tuned model (IT) as separate source models and compare them against all other models in the pool of 65, with results reported in Table 6. We omit the TinyLlama-1.1B families from Table 6, as they are trivial for every method. Each column corresponds to a single family (AUC computed over the corresponding tests with a family-calibrated threshold), whereas the overall score is computed across all test pairs under a single global threshold and is therefore *not* the average of the per-family scores. Three methods—*Intrinsic*, *ICS*, and our *SeedPrints*—are effectively saturated across every alteration (overall AUC of 0.994, 0.994, and 0.992; KS of 0.989, 0.943, and 0.986, respectively), attaining the maximal per-family score in most cases, whereas the remaining baselines, *REEF* and *Gradient*, trail substantially. The implication is striking: although *SeedPrints* is designed to capture *intrinsic seed-level* fingerprints, it simultaneously operates as a biometric-grade identifier that delivers reliable, persistent identity tracking across diverse model families—matching the best deployment-oriented baselines on their own terms while retaining a birth-stage guarantee that they cannot offer.

### 5.3.2 ALTERATION-WISE ROBUSTNESS

Across alteration types, *Instruct* is consistently the easiest case (even the weakest baseline, *Gradient*, reaches AUC 0.895), whereas *Finetune* and *Merge* are the most demanding overall—on *Finetune*, no method attains perfect performance. The remaining transformations yield a more mixed picture: *REEF* drops markedly under *Quantization* (0.996 → 0.871) and *Distillation* (0.996 → 0.858) while remaining resilient to *PEFT* (0.994), whereas *PEFT* and *Quantization* are the harder cases for *Gradient* (0.895 → 0.776/0.756). The three stronger methods are broadly robust across all transformations (every score > 0.9). *SeedPrints* declines only slightly, to 0.959, on *Merge*—a single misclassified merge model out of seventeen; this lone error indicates that weight-space interpolation with unrelated model weights can, in isolated cases, dilute the initialization signal toward the detection threshold. Averaged across all six alteration types, *Intrinsic* records the highest overall AUC, followed closely by *ICS* and *SeedPrints*—placing our method within a negligible margin of the best deployment-specialized baselines while additionally furnishing the birth-stage guarantees that they lack.

Table 7: Performance of LLM fingerprinting methods across parameter-altering techniques. SeedPrints remains robust across all six transformations ( $\text{AUC} \geq 0.959$ ), with its only appreciable dip on *Merge*, and attains an overall AUC/KS of 0.991/0.985—closely tracking the strongest baselines while additionally providing the birth-stage verification that they lack.

| Method     | Metric                  | Instruct | Finetune | PEFT  | Quantization | Merge | Distillation | Overall Avg  |
|------------|-------------------------|----------|----------|-------|--------------|-------|--------------|--------------|
| REEF       | AUC $\uparrow$          | 0.996    | 0.872    | 0.994 | 0.871        | 0.954 | 0.858        | <b>0.924</b> |
|            | KS Statistic $\uparrow$ | 0.949    | 0.637    | 0.940 | 0.812        | 0.851 | 0.615        | <b>0.801</b> |
| Gradient   | AUC $\uparrow$          | 0.895    | 0.729    | 0.776 | 0.756        | 0.832 | 0.887        | <b>0.812</b> |
|            | KS Statistic $\uparrow$ | 0.756    | 0.441    | 0.469 | 0.586        | 0.519 | 0.679        | <b>0.575</b> |
| ICS        | AUC $\uparrow$          | 1.000    | 0.987    | 1.000 | 1.000        | 0.981 | 1.000        | <b>0.995</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.925    | 1.000 | 1.000        | 0.941 | 1.000        | <b>0.978</b> |
| Intrinsic  | AUC $\uparrow$          | 1.000    | 0.980    | 1.000 | 1.000        | 1.000 | 1.000        | <b>0.997</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.971    | 1.000 | 1.000        | 1.000 | 1.000        | <b>0.995</b> |
| SeedPrints | AUC $\uparrow$          | 1.000    | 0.995    | 0.990 | 1.000        | 0.959 | 1.000        | <b>0.991</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.990    | 0.980 | 1.000        | 0.941 | 1.000        | <b>0.985</b> |

#### 5.4 ABLATION STUDY ON HYPERPARAMETERS

Our method exposes three hyperparameters: the number of random input sequences  $n$ , the length of each random sequence  $l$ , and the fingerprint size  $m$ . The first two jointly govern the total volume of random input, whereas  $m$  fixes how many low-mean coordinates enter the test. Throughout the main experiments in Sections 5.2 and 5.3 we adopt  $n=2000$ ,  $l=1024$ , and  $m=0.1d_{\text{out}}$ , where  $d_{\text{out}}$  is the hidden-state dimensionality—a specification that adapts automatically across open-source models of differing width. Here we vary each hyperparameter in turn while holding the others at their defaults ( $n = 2000$ ,  $m = 400$ ,  $l = 768$ ), reporting the empirical accuracy and false-positive rate at a fixed significance level  $\alpha = 0.01$  in Table 8.

Increasing either  $n$  or  $l$  consistently improves identification performance, exactly as our theoretical analysis in Appendix C.2 predicts: accurate recovery of the population-level bias requires the number of queries to satisfy  $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$ , where  $\gamma$  is the bias margin that separates the biased coordinates from the remainder. A given output dimension therefore demands a minimum sample count for reliable detection, which accounts for the degraded performance at small  $n$ . The margin  $\gamma$  in turn grows with  $l$  through accumulation effects, so longer sequences sharpen the separation; consequently, when the number of sequences is fixed and insufficient, increasing  $l$  lowers the effective query requirement and restores detection.

The effect of  $m$  is, by contrast, non-monotonic, in accordance with the theory. When  $m$  is too small, the selected coordinates are unstable, because too few empirical observations remain to form a reliable distribution; when  $m$  is too large, the many irrelevant coordinates admitted into the set inject noise that shrinks  $\gamma$ , which would in turn demand quadratically more samples to offset. Empirically,  $m$  in the range 200–400 performs well across model sizes, consistent with our heuristic  $m \approx 0.1d_{\text{out}}$  for the typical range  $d_{\text{out}} \in [3000, 5000]$ .

## 6 CONCLUSION

We introduced *SeedPrints*, a stronger and genuinely intrinsic notion of LLM fingerprinting that traces a model’s lineage all the way back to its random initialization. Our central finding is that initialization itself leaves a persistent “Galtonian” fingerprint on neural language models: untrained models already exhibit stable, seed-dependent token-preference patterns, and these biases persist—statistically and measurably—throughout training. Building directly on this finding, we developed a distribution-

Table 8: Hyperparameter ablation at significance level  $\alpha = 0.01$ : accuracy improves monotonically with larger  $n$  and  $l$ , whereas  $m$  exhibits the predicted non-monotonic behavior.

| Ablation on $n$ | 200    | 500    | 2000   |
|-----------------|--------|--------|--------|
| ACC             | 0.7368 | 0.7778 | 0.9375 |
| Empirical FPR   | 0      | 0      | 0      |
| Ablation on $l$ | 512    | 768    | 1024   |
| ACC             | 0.9048 | 0.9375 | 1.0000 |
| Empirical FPR   | 0.04   | 0      | 0      |
| Ablation on $m$ | 200    | 400    | 800    |
| ACC             | 0.9375 | 0.9375 | 0.875  |
| Empirical FPR   | 0.04   | 0      | 0.1463 |

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correlation test over *identity dimensions* that certifies shared lineage with calibrated statistical significance rather than through a heuristic threshold. Across LLaMA- and Qwen-style families and a broad suite of realistic deployments—continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation—SeedPrints delivers “birth-to-lifecycle” verification and withstands substantial domain shift, thereby unifying birth-stage and deployment-stage attribution within a single framework and furnishing a practical, reliable tool for provenance and copyright auditing.

**Limitations and Future Directions** Our objective in this work is to uncover fingerprint signals rooted in initialization-stage bias, and the method accordingly requires access to model outputs (logits or hidden states); it therefore may not apply when only restricted API access is available. We situate this boundary in context: the pursuit of *true fingerprints* for LLMs remains at an early stage, and most existing approaches—black-box ones in particular—depend on signals that surface only in later training or arise from alignment-driven artifacts, and consequently cannot robustly protect models at the pretrained stage. This state of affairs renders reliable all-stage fingerprints all the more urgently needed—precisely the gap our method begins to close. Full discussions appear in Appendix [A.2](#).

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## A LLM USAGE AND LIMITATION DISCUSSIONS

### A.1 LLM USAGE

We employed AI assistants, namely ChatGPT and Gemini, for writing and formatting support only; their use was confined to grammar and style checking, clarifying figure and table captions, and comparable surface-level edits. For programming tasks, we occasionally used GitHub Copilot and Claude as coding assistants—for instance, for code auto-completion and debugging hints. No AI assistant contributed to the scientific content, experimental design, or results reported in this paper.

### A.2 LIMITATIONS AND FUTURE DIRECTIONS

Although this work uncovers seed-born, training-persistent biases that serve as reliable fingerprint signals, the proposed method operates within the white-box paradigm and thus may not apply when only restricted API access is available—for instance, when responses are deterministic or safety-filtered. We situate this limitation within a broader observation: the pursuit of *true fingerprints* for LLMs remains at a very early stage even in the white-box setting, because most existing approaches rest on training-induced signatures or alignment behaviors and consequently cannot robustly protect pretrained or early-stage models. By analogy to human fingerprints, which remain stable irrespective of age or environment, a *true* LLM fingerprint should persist across continued training and deployment rather than merely reflect incidental training artifacts—which is precisely the criterion our method is expressly designed to satisfy. Extending this guarantee to the black-box setting, where a substantial performance gap relative to white-box methods still persists, is a challenging and important open direction. We hope this work catalyzes further research toward principled, robust fingerprinting frameworks for large models.

## B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

Our hypothesis test in Section 4 decides whether two models share lineage by comparing the distribution of their correlation statistics over random inputs against a null distribution—the distribution of correlation statistics between two *independent models evaluated on random inputs*.

We consider two approximations to this null: (i) an empirical, simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived in closed form from the known distribution of Kendall–Tau correlations. We justify both in three steps: we first characterize the true null via independently initialized models, then introduce the Gaussian surrogate as a tractable simulation-based proxy, and finally validate the analytical approximation against that surrogate.

### B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null corresponds to the correlation statistics between two independent models evaluated on random inputs, and the cleanest way to instantiate this independence is to compare two independently initialized models (distinct random seeds, no training). We estimate this empirical null by evaluating  $\sim 2500$  such pairs and computing Kendall–Tau correlations on 10,000 random inputs (as in Section 3). Figure 3 confirms that the true null is empirically a Gaussian centered at zero—precisely the form our analytical treatment assumes.

### B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Directly estimating the null from independently initialized models is computationally expensive, demanding access to many models and repeated evaluations. As a tractable alternative, we introduce a simulation-based null that uses Gaussian surrogates as proxies for model outputs under the null.

Two considerations justify this choice. First, approximating neural-network outputs with Gaussian distributions is a well-established practice in the auditing literature, supported by both empirical and theoretical evidence (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline depends only on relative ordering and selection operations (rank correlations and bottom- $m$  selection) and makes no use of semantic structure in the inputs; moreover, randomly initialized models possess independent

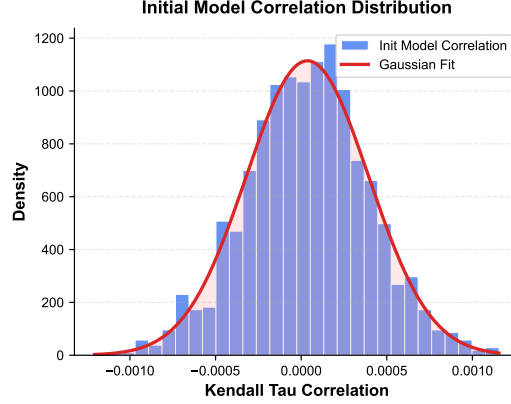


Figure 3: The empirical correlation distribution over  $\sim 2500$  pairs of independently initialized models, evaluated on random inputs, closely matches a Gaussian centered at zero, thereby directly validating the null model used in our test.

parameter matrices, so their outputs under the null exhibit no strong structured dependencies. Sampling independent Gaussian matrices is therefore a simple, faithful, and tractable proxy for the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape  $\mathbb{R}^{N \times d_{\text{out}}}$ ,
2. Extract the most biased dimensions by selecting the bottom- $m$  ranked average dimensions,
3. Take the intersection set  $\mathcal{S}$  and compute correlations on it.

Because this construction applies the identical selection procedure to Gaussian samples, it naturally captures the dependency patterns introduced by the pipeline itself (e.g., those arising from selecting  $\mathcal{S}$  across dimensions). As Figure 4 (middle) shows, the resulting distribution reproduces the centered Gaussian shape of the empirical null obtained from independently initialized models, thereby confirming that the Gaussian surrogate is a faithful simulation-based approximation of the true null.

### B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

Although the Gaussian surrogate faithfully captures the null under the full pipeline, it still entails repeated sampling and injects randomness into the estimation. We therefore derive a maximally efficient analytical approximation that eliminates sampling altogether while preserving this fidelity.

Under the null hypothesis that the two models are independent<sup>2</sup>, each Kendall–Tau correlation  $\tau_j$  is approximately zero-mean with known variance  $\sigma^2$ . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization does introduce dependencies across dimensions. Nonetheless, Figure 4 shows that the analytical Gaussian closely matches the distribution obtained from the Gaussian surrogate pipeline across a range of settings, including varying numbers of random inputs  $n$  and output dimensions  $D$ , which demonstrates that these dependencies are sufficiently weak to leave the approximation’s accuracy intact. The likely mechanism is that the relatively high temperature ( $T = 10$ ) smooths the output distribution and thereby suppresses inter-dimension coupling.

Together, these results justify the analytical Gaussian null adopted in Section 4 as an efficient, deterministic, and accurate approximation that obviates costly empirical baseline construction entirely.

<sup>2</sup>As discussed in Appendix B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

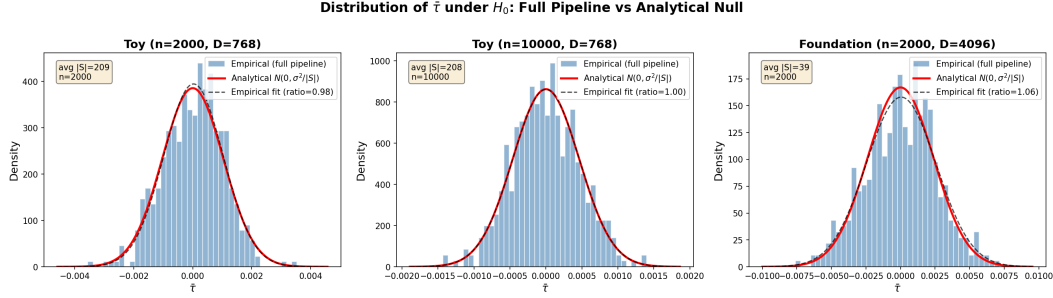


Figure 4: Validation of the analytical Gaussian null. Applying the full pipeline (including identity-dimension selection and correlation aggregation) to Gaussian surrogate outputs, we compare the resulting distribution of  $\bar{\tau}$  against the analytical Gaussian  $\mathcal{N}(0, \sigma^2/|S|)$ . The close agreement establishes that the analytical approximation accurately captures the null behavior in spite of the weak dependencies introduced by the pipeline.

**On variance scaling.** The true null obtained from randomly initialized models has slightly smaller variance than the Gaussian null approximation, yet this mismatch has no bearing on our hypothesis-test results. Our primary evaluation for the simulation-based null in Appendix D.2 uses the Mann–Whitney  $U$ -test, which is rank-based and hence invariant to any positive scaling: for any  $c > 0$ ,  $x_i < x_j \Leftrightarrow cx_i < cx_j$ , so every rank statistic and rejection threshold is left unchanged. For tests that do depend on absolute variance, we additionally evaluated variance-sensitive statistics such as the  $t$ -test, whose statistic scales as  $t' = t/c$  when the variance decreases. Because  $c < 1$  in our case, the scaling shrinks the magnitude of  $t$  and thus renders the test *more conservative* rather than inflating significance. Empirically, across the extensive experiments in Appendix D.2, this variance scaling leaves the power of the  $t$ -test intact, which indicates that the distributional separation between the alternative and the null is far larger than the scaled variance.

For the analytical null used in the main paper, the variance enters explicitly through the normalization of the  $z$ -score, so any scale discrepancy is absorbed into the denominator. Since the analytical variance is slightly larger than the empirical one, the resulting  $z$ -scores are correspondingly smaller, once again biasing the test toward conservatism and guarding against false positives.

## C THEORETICAL PROPERTIES OF OUR METHODS

Our goal is to identify the output dimensions that the initialized model is least likely to predict under random inputs. These dimensions correspond to directions that the model inherently disfavors owing to initialization-induced bias—a bias independent of the input distribution—and they form the basis of a stable fingerprint signal. This section formalizes the two empirical extractors, proves that the mean-based extractor we adopt is stable whereas per-sample voting is not, and characterizes the sample complexity that governs recovery.

**Setup.** Consider a model  $g$  with fixed parameters and random inputs  $x_i$  (uniform random tokens or random embeddings as in the main paper). For each output dimension  $j \in [d_{\text{out}}]$ ,  $g_j(x)$  is thus a random variable and its population mean is  $\bar{g}_j := \mathbb{E}[g_j(x)]$ . The population-level directions we aim to capture correspond to the  $m$  output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$  denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the  $m$ -th and  $(m+1)$ -th smallest means. This gap quantifies how well-separated the target set  $\mathcal{M}$  is from the remaining coordinates. Crucially, for a fixed  $m$  and input distribution,  $\Gamma(m)$  depends solely on the model’s inherent initialization bias. *Increasing the length of random input sequences amplifies this bias via concentration, rendering the separation more pronounced at the population level.*

Two natural empirical strategies present themselves for extracting such population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs  $X = \{x_i\}_{i=1}^n$  and select the  $m$  coordinates with the smallest value. Following the notation in Section 4, where  $g$  denotes the model (either the base model  $f$  or the suspect model  $f'$ ):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input  $x_i \in \mathbb{R}^{\ell \times d}$ , select the smallest-value coordinate and then take the  $m$  most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let  $p_j := \Pr(j^*(x) = j)$  denote the population probability that index  $j$  attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- $m$  set under voting.

Both strategies are intuitively reasonable, yet they behave very differently under perturbation. As we prove below, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and can flip under arbitrarily small perturbations, which renders it inherently noise-sensitive. This contrast constitutes the formal justification for our design choice to extract identity dimensions from the mean response vector.

## C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

### C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first establish that the sample-mean-based operator—which selects the  $m$  smallest coordinates of the empirical mean vector—is stable under small perturbations to model outputs. Such perturbations model the realistic variation in outputs from the *same* model, for instance numerical noise or evaluation under different precision modes.

Formally, we write the perturbed outputs as  $g'(x) = g(x) + \eta(x)$ , where  $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$  denotes the stochastic deviation. We assume that for each sample  $x_i$  and coordinate  $j$ , the random variables  $\eta_j(x_i)$  are independent across samples (not necessarily across coordinates) and  $\sigma$ -sub-Gaussian with  $\mathbb{E}[\eta_j(x_i)] = 0$ .<sup>3</sup>

**Lemma C.1** (Stability of sample-mean-based extraction). *Let  $\hat{g}$  and  $\hat{g}'$  be the empirical means over  $X = \{x_i\}_{i=1}^n$ , i.e.,  $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$  and  $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$ . Let  $\widehat{\Gamma}(m)$  denote the empirical gap at rank  $m$ . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

**Interpretation.** The lemma provides a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. This instability probability decays exponentially in the number of averaged samples  $n$ —a direct consequence of the variance reduction from  $\sigma^2$  per sample to  $\sigma^2/n$  in the empirical mean.

<sup>3</sup>This sub-Gaussian assumption is mild: bounded random variables are  $\sigma$ -sub-Gaussian (Hoeffding’s lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case  $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$  is a special case of this assumption.

*Proof.* We condition on the sample  $X = \{x_i\}_{i=1}^n$  throughout the proof, so that  $\hat{g}$  and  $\hat{\Gamma}(m)$  are fixed, and all probabilities are taken over the randomness of  $\{\eta(x_i)\}_{i=1}^n$  only.

For any coordinate  $j$ , by definition of  $\hat{g}'$ ,  $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$ . Each  $\eta_j(x_i)$  is mean-zero and  $\sigma$ -sub-Gaussian, so their average is mean-zero and  $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as  $1/n$  under averaging. Hence, for any  $\varepsilon > 0$ , by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than  $\varepsilon$ :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all  $d_{\text{out}}$  coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \right\}.$$

Equivalently, if  $\max_j |\hat{g}'_j - \hat{g}_j| < \hat{\Gamma}(m)/2$ , then the bottom- $m$  identity set remains unchanged.

Let  $j_{(1)}, \dots, j_{(d_{\text{out}})}$  be a permutation of  $\{1, \dots, d_{\text{out}}\}$  such that  $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$ . Recall that the empirical gap at rank  $m$  is  $\hat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$ . Denote  $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$  and  $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$ . For any  $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$  and any  $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$ ,  $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \hat{\Gamma}(m)$ . With  $|\hat{g}'_i - \hat{g}_i| < \hat{\Gamma}(m)/2$  and  $|\hat{g}'_\ell - \hat{g}_\ell| < \hat{\Gamma}(m)/2$ , we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\hat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\hat{\Gamma}(m)}{2}) \geq \hat{\Gamma}(m) - \hat{\Gamma}(m) = 0.$$

Thus no coordinate in  $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$  can become smaller than any coordinate in  $\widehat{\mathcal{M}}^{\text{mean}}(g)$  under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since  $\hat{\Gamma}(m)$  is fixed under conditioning on  $X$ , plugging  $\varepsilon = \hat{\Gamma}(m)/2$  into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\hat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

### C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

By contrast, per-sample voting applies a discontinuous winner-take-all operator to each input, so that arbitrarily small output perturbations can flip its decisions with non-vanishing probability. We retain the same stochastic perturbation model  $g'(x) = g(x) + \eta(x)$ .

**Lemma C.2** (Instability of per-sample argmin voting). *For a fixed input  $x$ , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

*denote the argmin and its margin. Then, for any fixed  $x$ ,*

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

*and for  $n$  i.i.d. inputs  $x_1, \dots, x_n$ ,*

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

*where  $\alpha := \mathbb{E}_x \left[ \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$ .*

**Interpretation.** Unlike mean extraction, voting derives no benefit from variance reduction: each input incurs a hard decision made directly on noisy outputs. Increasing  $n$  therefore merely multiplies the opportunities for index-flips.

*Proof.* We first bound the per-input change probability  $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$ . Fix  $x$  and write  $j^* := j^*(x; g)$  for brevity. By definition of the margin,  $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$ . Let  $j^{(2)}$  be an index attaining this minimum, i.e.,  $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$ .

Under the perturbation  $g'(x) = g(x) + \eta(x)$ , we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since  $\eta_{j^{(2)}}(x)$  and  $\eta_{j^*}(x)$  are independent  $\mathcal{N}(0, \sigma^2)$  variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes,  $j^*(x; g') \neq j^*(x; g)$ , certainly occurs whenever  $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$ . Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let  $Z \sim \mathcal{N}(0, 1)$ . Since  $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$ , we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore,  $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$ .

Now consider  $n$  i.i.d. inputs  $x_1, \dots, x_n$ . Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across  $i$ , because the perturbations  $\eta(x_i)$  are independent across  $i$ . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define  $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$ , and  $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$ . Taking expectations over the randomness of  $x_i$ , we obtain  $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$ . Therefore, by independence across  $i$ ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function  $u \mapsto \log(1 - u)$  on  $u \in [0, 1)$ . Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set  $\widehat{\mathcal{M}}^{\text{vote}}$  is determined by the empirical counts of the dimensions  $\{j^*(x_i; g)\}_{i=1}^n$ . If the empirical gap between the  $m$ -th and  $(m+1)$ -th most frequent dimensions is of order  $O(1/n)$  (i.e., only a constant number of votes), then a change in  $j^*(x_i)$  for a single input  $x_i$  is sufficient to swap the ordering across this boundary and thus alter the top- $m$  set. Combining this observation with the lower bound  $1 - (1 - \alpha)^n$  shows that, in such regimes, the probability that  $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$  is bounded away from zero uniformly in  $n$ , i.e., per-sample argmin voting does not enjoy variance reduction with more samples.  $\square$

## C.2 INFLUENCE OF HYPERPARAMETERS

We now quantify how many random inputs the empirical mean estimator requires in order to recover the true bias. Reusing the notation from **Setup**, each output coordinate  $g_j(x)$  is a random variable under random inputs  $x \sim \mathcal{D}$ , with population mean  $\bar{g}_j := \mathbb{E}[g_j(x)]$ .  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$  represents the sorted means in non-decreasing order, and  $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$  is the separation margin. We estimate  $\bar{g}$  using  $n$  i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimensions set is  $\mathcal{M}$  with size  $m$  and our empirical estimation gives  $\widehat{\mathcal{M}}$ .

**Lemma C.3** (Sufficient condition for recovering the true set). *If  $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$  and  $2\varepsilon < \gamma$ , then the empirical bottom- $m$  set equals the true set:  $\widehat{\mathcal{M}} = \mathcal{M}$ .*

*Proof.* For any  $j \in \mathcal{M}$  and  $j' \notin \mathcal{M}$ , we have  $\bar{g}_j \leq \bar{g}_{(m)}$  and  $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$ . The  $\ell_\infty$  bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- $m$  set.  $\square$

We next translate this recovery condition into hyperparameter terms, adopting the same sub-Gaussian assumption as in Appendix C.1. The assumption is mild: by Hoeffding's inequality, any bounded random variable is  $\sigma$ -sub-Gaussian with  $\sigma$  proportional to its range, and in our setting each  $g_j(x)$  is a logit or a final hidden activation—quantities that are typically bounded or light-tailed in both initialized networks (owing to controlled parameter initialization) and trained networks.

**Corollary C.4** (Sample complexity under sub-Gaussian coordinates). *Suppose each  $g_j(x)$  is  $\sigma$ -sub-Gaussian. Then with probability at least  $1 - \delta$ ,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then  $\widehat{\mathcal{M}} = \mathcal{M}$  with probability at least  $1 - \delta$ .

*Proof.* For fixed  $j$ , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all  $d_{\text{out}}$  coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to  $\delta$  and solve for  $\varepsilon$  to obtain, with probability at least  $1 - \delta$ ,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If  $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$  (i.e., if  $2\varepsilon < \gamma$ ), then Lemma C.3 applies and implies  $\widehat{\mathcal{M}} = \mathcal{M}$  with probability at least  $1 - \delta$ .  $\square$

Table 9: Fingerprint behavior under different random initializations across model families: every distinct-seed pair yields  $p > 0.01$  under both tests and both output types, so SeedPrints separates seeds reliably in every configuration examined.

| (a) LLaMA-style models    |                |                |                |                | (b) Qwen-style models    |                |                |                |                |
|---------------------------|----------------|----------------|----------------|----------------|--------------------------|----------------|----------------|----------------|----------------|
| Seed Pair                 | Logits Output  |                | Hidden State   |                | Seed Pair                | Logits Output  |                | Hidden State   |                |
|                           | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |                          | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |
| $s_{42}$ vs. $s_{2000}$   | 0.404          | 0.456          | 0.357          | 0.532          | $s_{123}$ vs. $s_{1000}$ | 0.741          | 0.727          | 0.094          | 0.074          |
| $s_{123}$ vs. $s_{42}$    | 0.214          | 0.295          | 0.678          | 0.565          | $s_{1000}$ vs. $s_{123}$ | 0.954          | 0.971          | 0.125          | 0.094          |
| $s_{1000}$ vs. $s_{123}$  | 0.219          | 0.246          | 0.363          | 0.335          | $s_{42}$ vs. $s_{2000}$  | 0.273          | 0.360          | 0.451          | 0.529          |
| $s_{2000}$ vs. $s_{1000}$ | 0.282          | 0.291          | 0.434          | 0.481          | $s_{2000}$ vs. $s_{42}$  | 0.215          | 0.206          | 0.230          | 0.295          |

Table 10: (LLaMA-style models) Trained models retain the fingerprint of their initialization ( $p$ -value  $< 0.01$  across both output types and both tests), whereas not one of the four baselines detects the shared lineage in any configuration.

| Model Pair                              | Logits Output  |                | Hidden State   |                | Baselines |        |        |        |
|-----------------------------------------|----------------|----------------|----------------|----------------|-----------|--------|--------|--------|
|                                         | <i>t</i> -test | <i>u</i> -test | <i>t</i> -test | <i>u</i> -test | Intrinsic | REEF   | PCS    | ICS    |
| $s_{42}^{init}$ vs. $s_{42}^{base}$     | 3.33e-3✓       | 1.02e-3✓       | 2.20e-8✓       | 6.28e-8✓       | -0.021×   | 0.375× | 0.580× | 0.196× |
| $s_{123}^{init}$ vs. $s_{123}^{base}$   | 2.06e-3✓       | 7.33e-3✓       | 7.09e-6✓       | 1.37e-5✓       | 0.149×    | 0.369× | 0.581× | 0.188× |
| $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 2.44e-3✓       | 4.14e-3✓       | 5.58e-4✓       | 2.81e-3✓       | -0.252×   | 0.381× | 0.581× | 0.188× |
| $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 5.63e-3✓       | 6.76e-3✓       | 4.00e-10✓      | 1.27e-9✓       | -0.337×   | 0.331× | 0.581× | 0.188× |

Consequently, accurate recovery requires  $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$ : the number of random queries grows only logarithmically in the output dimension yet scales inversely with the squared separation margin  $\gamma$ . The margin  $\gamma$  is in turn governed by two factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias through concentration, thereby widening the population separation between coordinates.
- **Choice of  $m$ .** Different values of  $m$  probe different regions of the ranked spectrum  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$  and thus induce different effective margins in practice.

The former is controlled through the input length  $l$ , whereas the latter is selected empirically from the observed gaps in the mean spectrum.

## D MORE EXPERIMENT DETAILS

### D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models with the Hugging Face Transformers Trainer (Wolf et al., 2020) and employ Accelerate (Gugger et al., 2022) for distributed runs. All open-source models are loaded from their official Hugging Face releases and used under their original licenses—Llama models under the Meta Llama Community License and the remainder under Apache-2.0—and all datasets are obtained through the Hugging Face Datasets library (itself Apache-2.0), with dataset content governed by each dataset’s stated license.

### D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section provides the complementary results for Section 5. We report results using the empirical simulation-based null of Algorithm 1 with both *logits* and *hidden state* outputs, evaluated under both the one-sided *t*-test (*t*-test) and the Mann–Whitney *U* test (*U* test); across every one of these configurations, the conclusions of the main paper are reproduced without exception, which establishes that our findings are robust to the choice of null construction, output representation, and test statistic.

Table 11: (Qwen-style models) Trained models retain the fingerprint of their initialization ( $p$ -value  $< 0.01$  across both output types and both tests), corroborating the LLaMA-style results and establishing that the effect is architecture-independent.

| Model Pair                              | Logits Output |           | Hidden State |           |
|-----------------------------------------|---------------|-----------|--------------|-----------|
|                                         | $t$ -test     | $U$ -test | $t$ -test    | $U$ -test |
| $s_{123}^{init}$ vs. $s_{123}^{base}$   | 2.38e-03      | 1.68e-03  | 7.36e-15     | 3.38e-13  |
| $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 1.35e-04      | 4.84e-05  | 4.41e-13     | 2.01e-11  |
| $s_{42}^{init}$ vs. $s_{42}^{base}$     | 1.39e-03      | 1.46e-03  | 1.06e-24     | 2.05e-19  |
| $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 1.80e-03      | 1.28e-03  | 4.87e-24     | 1.92e-20  |

Table 12: Fingerprint persistence under continual training on diverse datasets (base model: seed 1000, corpus openwebtext); the  $U$ -test denotes the Mann–Whitney  $U$  test. SeedPrints attributes lineage correctly on both TinyStories and the sharply shifted code corpus (The Stack) under logits and hidden states alike, whereas the baselines fail on The Stack, corroborating the main-paper finding under every test and output type.

(a) LLaMA-style models

| Setting                 | Ours (logits)      |                        | Ours (hidden)      |                       | Baselines          |                    |                    |                    |
|-------------------------|--------------------|------------------------|--------------------|-----------------------|--------------------|--------------------|--------------------|--------------------|
| Continual corpus (seed) | $t$ -test          | $U$ -test              | $t$ -test          | $U$ -test             | Intrinsic          | REEF               | PCS                | ICS                |
| TinyStories (1000)      | 0 <sup>✓</sup>     | 0 <sup>✓</sup>         | 0 <sup>✓</sup>     | 7.77e-89 <sup>✓</sup> | 1.000 <sup>✓</sup> | 0.759 <sup>×</sup> | 0.999 <sup>✓</sup> | 0.996 <sup>✓</sup> |
| TinyStories (123)       | 1.000 <sup>✓</sup> | 1.00 <sup>✓</sup>      | 0.943 <sup>✓</sup> | 0.902 <sup>✓</sup>    | 0.950 <sup>×</sup> | 0.658 <sup>×</sup> | 0.332 <sup>✓</sup> | 0.012 <sup>✓</sup> |
| the_stack (1000)        | 0 <sup>✓</sup>     | 1.73e-287 <sup>✓</sup> | 0 <sup>✓</sup>     | 3.09e-69 <sup>✓</sup> | 0.489 <sup>×</sup> | 0.557 <sup>×</sup> | 0.585 <sup>×</sup> | 0.123 <sup>×</sup> |
| the_stack (123)         | 0.616 <sup>✓</sup> | 0.479 <sup>✓</sup>     | 0.732 <sup>✓</sup> | 0.831 <sup>✓</sup>    | 0.445 <sup>✓</sup> | 0.580 <sup>×</sup> | 0.301 <sup>✓</sup> | 0.026 <sup>✓</sup> |

(b) Qwen-style models

| Setting                 | Ours (logits)         |                        | Ours (hidden)          |                       | Baselines          |                    |                    |                    |
|-------------------------|-----------------------|------------------------|------------------------|-----------------------|--------------------|--------------------|--------------------|--------------------|
| Continual corpus (seed) | $t$ -test             | $U$ -test              | $t$ -test              | $U$ -test             | Intrinsic          | REEF               | PCS                | ICS                |
| TinyStories (1000)      | 5.36e-09 <sup>✓</sup> | 1.92e-07 <sup>✓</sup>  | 8.49e-214 <sup>✓</sup> | 5.09e-71 <sup>✓</sup> | 1.000 <sup>✓</sup> | 0.957 <sup>✓</sup> | 0.999 <sup>✓</sup> | 0.996 <sup>✓</sup> |
| TinyStories (123)       | 0.434 <sup>✓</sup>    | 0.433 <sup>✓</sup>     | 0.256 <sup>✓</sup>     | 0.065 <sup>✓</sup>    | 0.913 <sup>×</sup> | 0.199 <sup>✓</sup> | 0.328 <sup>✓</sup> | 0.039 <sup>✓</sup> |
| the_stack (1000)        | 0 <sup>✓</sup>        | 4.16e-237 <sup>✓</sup> | 1.16e-211 <sup>✓</sup> | 2.30e-76 <sup>✓</sup> | 0.999 <sup>✓</sup> | 0.313 <sup>×</sup> | 0.995 <sup>✓</sup> | 0.976 <sup>✓</sup> |
| the_stack (123)         | 0.999 <sup>✓</sup>    | 0.993 <sup>✓</sup>     | 0.610 <sup>✓</sup>     | 0.491 <sup>✓</sup>    | 0.916 <sup>×</sup> | 0.255 <sup>✓</sup> | 0.328 <sup>✓</sup> | 0.038 <sup>✓</sup> |

## D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based random baseline is stochastic, we report  $p$ -values averaged over 10 independent trials at a significance level of  $\alpha = 0.01$ . We emphasize that the absolute magnitude of extremely small  $p$ -values carries no additional meaning: once  $p$  falls below the floor set by numerical and sampling noise (e.g.,  $< 10^{-20}$ ), a value such as  $10^{-260}$  constitutes no stronger evidence than  $10^{-20}$ , since both decisively reject the null.

**Different initialization seeds produce distinct fingerprints.** Table 9a and Table 9b report  $p$ -values from our correlation test between pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under the  $t$ -test and  $U$ -test, for LLaMA-style and Qwen-style models respectively. Every  $p$ -value exceeds 0.01, so our method reliably separates models originating from different seeds: distinct seeds yield distinct fingerprint behaviors, and the models remain separable “at birth” under both tests and both architectures.

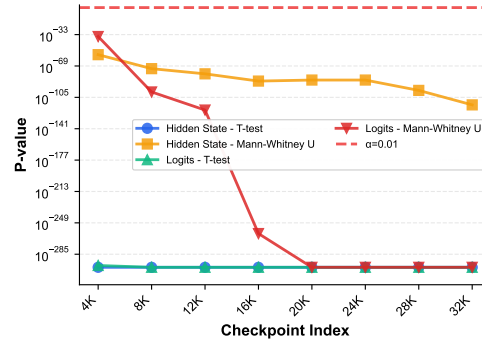
**Training preserves the initialization fingerprint.** Table 10 (LLaMA-style) and Table 11 (Qwen-style) compare each initialization model  $s^{init}$  with its descendant  $s^{base}$  trained on OpenWebText (Gokaslan et al., 2019) ( $\approx 10$ B tokens). Across every seed–model pair the  $p$ -values are uniformly  $< 0.01$ : the bias profiles remain strongly correlated and share a common lineage, so the trained model demonstrably inherits the fingerprint of its initialization, and training does not erase it. The

Table 13: Identical data and training order do not force fingerprint behaviors to coincide across different initializations: every cross-seed pair yields  $p > 0.01$ , under both output types and both tests.

| (a) LLaMA-style models                  |               |           |              |           | (b) Qwen-style models                  |               |           |              |           |
|-----------------------------------------|---------------|-----------|--------------|-----------|----------------------------------------|---------------|-----------|--------------|-----------|
| Model Pair                              | Logits Output |           | Hidden State |           | Model Pair                             | Logits Output |           | Hidden State |           |
|                                         | $t$ -test     | $U$ -test | $t$ -test    | $U$ -test |                                        | $t$ -test     | $U$ -test | $t$ -test    | $U$ -test |
| $s_{123}^{init}$ vs. $s_{1000}^{base}$  | 0.484         | 0.500     | 0.385        | 0.486     | $s_{123}^{init}$ vs. $s_{1000}^{base}$ | 0.598         | 0.638     | 0.286        | 0.254     |
| $s_{1000}^{init}$ vs. $s_{2000}^{base}$ | 0.946         | 0.956     | 0.135        | 0.096     | $s_{1000}^{init}$ vs. $s_{123}^{base}$ | 0.804         | 0.805     | 0.236        | 0.240     |
| $s_{42}^{init}$ vs. $s_{123}^{base}$    | 0.598         | 0.589     | 0.426        | 0.337     | $s_{42}^{init}$ vs. $s_{2000}^{base}$  | 0.589         | 0.608     | 0.226        | 0.2043    |
| $s_{2000}^{init}$ vs. $s_{42}^{base}$   | 0.756         | 0.781     | 0.388        | 0.287     | $s_{2000}^{init}$ vs. $s_{42}^{base}$  | 0.523         | 0.482     | 0.312        | 0.323     |



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: SeedPrints verifies lineage at every checkpoint ( $p$ -values  $< 0.01$ ), doing so consistently across both model families.

baselines in Table 10 behave in exactly the opposite way: without exception they fail to distinguish across seeds, which confirms that their separability originates in training-induced artifacts rather than in initialization.

**Identical data and order do not force fingerprints to converge.** In Table 13a (LLaMA-style) and Table 13b (Qwen-style), all four “suspicious” models  $s_i^{base}$  for  $i \in \{42, 123, 100, 2000\}$  are trained on *exactly the same corpus (OpenWebText) and in the same data order*—we fix the training seed to lock the curriculum—so that their sole difference is the initialization seed  $i$ . The aim is to test whether identical data and order can erase or confound the fingerprint. Across every cross-seed pair the  $p$ -values remain  $> 0.01$ , in sharp contrast to the near-zero values of Table 10: the fingerprint stays seed-specific even under identical data and curriculum.

**Continual training on diverse datasets does not confound the fingerprint.** The experiments above serve to establish that SeedPrints behaves as a biometric fingerprint. From a copyright standpoint, the inability of prior methods to separate models by seed is not in itself a shortcoming, since initialization seeds carry no clear copyright status; their genuine fragility lies elsewhere—their attribution is easily misled by *data distribution*, *failing to recognize lineage when the training distribution shifts substantially during continued training*.

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two deliberately dissimilar corpora: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We contrast (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (initialized with seed 123 and trained with a different data order on OpenWebText) and then continually trained on the *same* corpus, so that the decisive question becomes whether an attribution method can identify which descendant genuinely shares lineage with the base.

Every prior baseline fails in the code setting (The Stack), misclassifying true descendants as distractors—direct evidence that these methods track *domain* similarity rather than lineage identity: TinyStories remains close in distribution to the pretraining corpus (OpenWebText), whereas The

Table 14: Cross-size similarity between Qwen-2.5-14B (target) and other Qwen-2.5 models. SeedPrint reports  $p$ -values (higher is desirable here, since none of the smaller models is a true descendant), whereas REEF and HUREF report similarity scores.

|                          | Qwen-2.5-0.5B          | Qwen-2.5-1.5B          | Qwen-2.5-3B            | Qwen-2.5-7B            |
|--------------------------|------------------------|------------------------|------------------------|------------------------|
| SeedPrint ( $\uparrow$ ) | 0.7184                 | 0.9707                 | 0.3898                 | 0.5429                 |
| REEF ( $\downarrow$ )    | 0.8250                 | 0.8399                 | 0.8833                 | 0.8461                 |
| HUREF ( $\downarrow$ )   | $-6.37 \times 10^{-5}$ | $-2.84 \times 10^{-5}$ | $-8.70 \times 10^{-6}$ | $-6.08 \times 10^{-6}$ |

Stack diverges sharply, and such a pronounced shift alone suffices to defeat detection. Our method, by contrast, attributes lineage correctly under both corpora and both architectures. SeedPrints is therefore not a proxy for data distribution: it withstands substantial domain shift and persists well beyond the initial pretraining stage.

## D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

**Controlled pretraining trajectory.** We first conduct a controlled pretraining experiment on OpenWebText, in which the initialization seed is known. Treating each intermediate checkpoint as the base, we test whether our method reliably identifies its offspring along the same training trajectory. Every variant consistently recognizes the suspect model as same-lineage, with  $p$ -values remaining below the 0.01 threshold throughout (Figure 5a for LLaMA-style models and Figure 5b for Qwen-style models), which confirms the stability of SeedPrints under continuous optimization across both architectures.

## D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To examine whether SeedPrints “collapses” into a single family-level signature when the parameter size is varied, we report cross-size similarity between Qwen-2.5-14B (target) and smaller models drawn from the same Qwen-2.5 family. Table 14 presents the results for SeedPrint ( $\uparrow$ ) alongside two baselines, REEF ( $\downarrow$ ) and HUREF ( $\downarrow$ ).

Concretely, to test whether initialization-born fingerprints collapse to a family-level signature under scaling, we conduct a cross-size study within the Qwen-2.5 series: we treat Qwen-2.5-14B as the protected target model and Qwen-2.5-{0.5B, 1.5B, 3B, 7B} as suspect models that share the architectural family yet are not trained as descendants of Qwen-2.5-14B. As established in the main paper, SeedPrints returns a  $p$ -value for the hypothesis “the suspect shares initialization-born fingerprints with the target,” whereas REEF and HUREF are similarity-based baselines that emit continuous scores and therefore require the user to fix a threshold.

Several observations stand out. First, every SeedPrints  $p$ -value lies comfortably above a typical significance level (e.g.,  $\alpha=0.01$ ), so under our hypothesis-testing view none of the smaller Qwen-2.5 models is flagged as sharing lineage with Qwen-2.5-14B. This is precisely the desired behavior: even though these models belong to the same family and reuse a similar architectural design, the test does not collapse to a single family-level signature that would spuriously conflate distinct pretraining runs.

Second, the cross-size  $p$ -values populate different regions of  $[0, 1]$  rather than clustering around a single value. We do *not* read these magnitudes as a continuous “similarity score”—their role is to support a calibrated yes/no decision at a fixed significance level—yet the spread confirms that our test statistic remains sensitive to the specific combination of architecture and initialization seed, rather than degenerating into a generic pattern common to all Qwen-2.5 variants.

Finally, the similarity-based baselines behave markedly differently. REEF produces scores in a narrow band (0.8250–0.8833) across all sizes, and HUREF outputs values numerically close to zero (on the order of  $10^{-5}$ ). Because neither method supplies a built-in decision rule, operators must hand-tune thresholds, and it is unclear how any single cutoff could robustly separate cross-size variants from genuinely lineage-related models. SeedPrints, by contrast, yields a statistical decision at a prescribed significance level, and thereby furnishes a clearer and more operational answer to the question “do these two models plausibly share a training lineage?” within a model family.

## E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers **58** models in total. Below we document the decoding configuration used throughout all runs and enumerate the model catalog grouped by family; for each derived model we additionally record the post-training transformation applied (*finetune*, *adapter*, *merge*, *quantization*, or *distillation*). Throughout, “PT” and “IT” denote *pre-trained* and *instruction-tuned* source models, respectively. The catalog below documents this coverage in full, enabling exact reproduction of every reported comparison.

| Parameter        | Value |
|------------------|-------|
| max_new_tokens   | 512   |
| temperature      | 0.7   |
| top_p            | 0.9   |
| top_k            | 50    |
| do_sample        | true  |
| max_input_length | 512   |

### E.1 MODEL LIST BY FAMILY

#### E.1.1 QWEN-2.5-7B

| Variant | HuggingFace repo                             | Type         |
|---------|----------------------------------------------|--------------|
| PT      | Qwen/Qwen2.5-7B                              | —            |
| IT      | Qwen/Qwen2.5-7B-Instruct                     | —            |
| Derived | Qwen/Qwen2.5-Math-7B                         | finetune     |
| Derived | Qwen/Qwen2.5-Coder-7B-Instruct               | finetune     |
| Derived | WangCa/Qwen2.5-7B-Medicine                   | finetune     |
| Derived | huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2 | finetune     |
| Derived | Locutusque/StockQwen-2.5-7B                  | merge        |
| Derived | bunnycore/QevaCoT-7B-Stock                   | merge        |
| Derived | fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct | adapter      |
| Derived | SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct    | adapter      |
| Derived | Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4           | quantization |
| Derived | Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8           | quantization |
| Derived | Lansechen/Qwen2.5-7B-Open-R1-Distill         | distillation |

#### E.1.2 QWEN2.5-14B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT      | Qwen/Qwen2.5-14B                                | —            |
| IT      | Qwen/Qwen2.5-14B-Instruct                       | —            |
| Derived | Qwen/Qwen2.5-Coder-14B                          | finetune     |
| Derived | oxyapi/oxy-1-small                              | finetune     |
| Derived | v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno | merge        |
| Derived | ToastyPigeon/qwen-story-test-qlora              | adapter      |
| Derived | Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4             | quantization |
| Derived | deepseek-ai/DeepSeek-R1-Distill-Qwen-14B        | distillation |

### E.1.3 LLAMA-3.1-8B

| Variant | HuggingFace repo                                                 | Type         |
|---------|------------------------------------------------------------------|--------------|
| PT      | meta-llama/Llama-3.1-8B                                          | —            |
| IT      | meta-llama/Llama-3.1-8B-Instruct                                 | —            |
| Derived | ValiantLabs/Llama3.1-8B-Fireplace2                               | finetune     |
| Derived | RedHatAI/Llama-3.1-8B-tldr                                       | finetune     |
| Derived | proxectonos/Llama-3.1-Carballo                                   | finetune     |
| Derived | mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated                  | finetune     |
| Derived | gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES       | merge        |
| Derived | Xiaojian9992024/Llama3.1-8B-ExtraMix                             | merge        |
| Derived | LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching | adapter      |
| Derived | chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7                     | adapter      |
| Derived | iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8                   | quantization |
| Derived | DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ                            | quantization |
| Derived | asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill                    | distillation |

### E.1.4 MISTRAL-7B-v0.3

| Variant | HuggingFace repo                                    | Type         |
|---------|-----------------------------------------------------|--------------|
| PT      | mistralai/Mistral-7B-v0.3                           | —            |
| IT      | mistralai/Mistral-7B-Instruct-v0.3                  | —            |
| Derived | KurmaAI/AQUA-7B                                     | finetune     |
| Derived | openfoodfacts/spellcheck-mistral-7b                 | finetune     |
| Derived | grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B      | merge        |
| Derived | chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer     | adapter      |
| Derived | RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit         | quantization |
| Derived | eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b | distillation |

### E.1.5 GEMMA-2-2B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT      | google/gemma-2-2b                               | —            |
| IT      | google/gemma-2-2b-it                            | —            |
| Derived | rinna/gemma-2-baku-2b                           | finetune     |
| Derived | anakin87/gemma-2-2b-neogenesis-ita              | finetune     |
| Derived | vonjack/gemma2-2b-merged                        | merge        |
| Derived | google-cloud-partnership/gemma-2-2b-it-lora-sql | adapter      |
| Derived | qilowoq/gemma-2-2B-it-4Bit-GPTQ                 | quantization |
| Derived | Syed-Hasan-8503/Gemma-2-2b-it-distilled         | distillation |

### E.1.6 LLAMA-2-7B

| Variant | HuggingFace repo                                                | Type         |
|---------|-----------------------------------------------------------------|--------------|
| PT      | meta-llama/Llama-2-7b-hf                                        | —            |
| IT      | meta-llama/Llama-2-7b-chat-hf                                   | —            |
| Derived | allenai/tulu-2-7b                                               | finetune     |
| Derived | QIAIUNCC/EYE-Llama_qa                                           | finetune     |
| Derived | DevQuasar/coma-7B-v0.1                                          | merge        |
| Derived | Ammar-1/llama2-Better-Tune                                      | adapter      |
| Derived | TheBloke/Llama-2-7B-Chat-GPTQ                                   | quantization |
| Derived | cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2 | distillation |

### E.1.7 TINYLLAMA-1.1B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT/IT   | TinyLlama/TinyLlama-1.1B-Chat-v1.0              | —            |
| Derived | alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2 | finetune     |
| Derived | Edentns/DataVortexTL-1.1B-v0.1                  | finetune     |
| Derived | appvoid/dot-v2.7                                | merge        |
| Derived | barissgic/tinyllama-tarot-v1                    | adapter      |
| Derived | TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ          | quantization |
| Derived | anudaw/distilled-code-llama                     | distillation |

**Notes.** “Type” denotes the post-training transformation applied relative to the PT/IT source model: *finetune* spans supervised and preference-optimization variants; *adapter* spans LoRA/QLoRA-style modules; *merge* spans model soups and TIES-style merges; *quantization* spans GPTQ/INTx variants; and *distillation* spans student models distilled from reasoning-augmented teachers. Together these categories span the full spectrum of parameter-altering transformations encountered in practical deployment.

## E.2 SAMPLE RESULTS

We present partial results from a single LeafBench evaluation run in Table 15 (*u*-test with the simulation-based null); the reported metric is  $1 - p$ , used as a similarity score. We fix the significance level for *p*-values at 0.01; empirically, LeafBench identifies an optimal decision threshold of 0.9920, which further corroborates the reliability of our method.

The distributional structure is what matters here: although individual *p*-values should not be read as linear similarity measures, their distribution across model pairs is highly informative—same-lineage models yield markedly lower *p*-values, whereas unrelated models remain approximately uniform, which is exactly the separation that a calibrated test is designed to exploit.

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| Model                                                      | Qwen-2.5-7B | Qwen-2.5-7B-Instruct | Qwen2.5-14B | Qwen2.5-14B-Instruct |
|------------------------------------------------------------|-------------|----------------------|-------------|----------------------|
| Qwen-2.5-7B                                                | 1.000       | 1.000                | 0.079       | 0.000                |
| Qwen-2.5-7B-Instruct                                       | 1.000       | 1.000                | 0.813       | 0.536                |
| Qwen2.5-7B-Math                                            | 1.000       | 0.994                | 0.047       | 0.987                |
| Qwen2.5-7B-Coder                                           | 0.996       | 0.999                | 0.177       | 0.002                |
| Qwen2.5-7B-Instruct-Medicine                               | 1.000       | 1.000                | 0.576       | 0.369                |
| Qwen2.5-7B-Instruct-Abiliterated                           | 1.000       | 1.000                | 0.276       | 0.886                |
| Qwen2.5-7B-Stock                                           | 1.000       | 1.000                | 0.939       | 0.225                |
| QevaCoT-7B                                                 | 1.000       | 1.000                | 0.199       | 0.119                |
| Qwen2.5-7B-Instruct-Task-10                                | 1.000       | 1.000                | 0.672       | 0.090                |
| Qwen2.5-7B-Instruct-Task-12                                | 1.000       | 1.000                | 0.155       | 0.987                |
| Qwen2.5-7B-Instruct-Int4                                   | 1.000       | 1.000                | 0.827       | 0.961                |
| Qwen2.5-7B-Instruct-Int8                                   | 1.000       | 1.000                | 0.914       | 0.739                |
| Qwen2.5-7B-Open-R1-Distill                                 | 1.000       | 1.000                | 0.302       | 0.955                |
| Qwen2.5-14B                                                | 0.324       | 0.487                | 1.000       | 1.000                |
| Qwen2.5-14B-Instruct                                       | 0.403       | 0.702                | 1.000       | 1.000                |
| Qwen2.5-Coder-14B                                          | 0.492       | 0.660                | 1.000       | 1.000                |
| oxy-1-small                                                | 0.492       | 0.104                | 1.000       | 1.000                |
| Qwen2.5-14B-Gutenberg-Instruct-Slerpeno                    | 0.856       | 0.931                | 1.000       | 1.000                |
| Qwen-story-test-qlora                                      | 0.112       | 0.898                | 1.000       | 1.000                |
| Qwen2.5-14B-Instruct-GPTQ-Int4                             | 0.583       | 0.092                | 1.000       | 1.000                |
| DeepSeek-R1-Distill-Qwen-14B                               | 0.204       | 0.000                | 1.000       | 1.000                |
| Llama-3.1-8B                                               | 0.638       | 0.714                | 0.156       | 0.000                |
| Llama-3.1-8B-Instruct                                      | 0.643       | 0.118                | 0.128       | 0.510                |
| Llama-3.1-8B-Fireplace2                                    | 0.715       | 0.532                | 0.065       | 0.354                |
| Llama-3.1-8B-TLDR                                          | 0.925       | 0.811                | 0.331       | 0.014                |
| Llama-3.1-8B-Carballo                                      | 0.001       | 0.605                | 0.003       | 0.946                |
| Llama-3.1-8B-Instruct-Abliterated                          | 0.061       | 0.005                | 0.121       | 0.422                |
| Llama-3.1-8B-Instruct-HalfAbliterated-TIES                 | 0.356       | 0.509                | 0.796       | 0.000                |
| Llama-3.1-8B-ExtraMix                                      | 0.640       | 0.160                | 0.721       | 0.875                |
| Llama-3.1-8B-Instruct-cv-job-description-matching          | 0.684       | 0.292                | 0.004       | 0.001                |
| Llama-3.1-8B-Instruct-PsyCourse-fold7                      | 0.682       | 0.057                | 0.063       | 0.191                |
| Llama-3.1-8B-Instruct-8bit                                 | 0.798       | 0.214                | 0.154       | 0.385                |
| Llama-3.1-8B-Instruct-4bit                                 | 0.513       | 0.023                | 0.005       | 0.933                |
| Llama-3.1-8B-Instruct-Open-R1-Distill                      | 0.800       | 0.101                | 0.772       | 0.024                |
| Mistral-7B-v0.3                                            | 0.224       | 0.352                | 0.506       | 0.044                |
| Mistral-7B-v0.3-Instruct                                   | 0.000       | 0.000                | 0.984       | 0.104                |
| AQUA-7B                                                    | 0.014       | 0.613                | 0.703       | 0.569                |
| Mistral-7B-v0.3-Spellcheck                                 | 0.005       | 0.481                | 0.872       | 0.105                |
| Mistral-7B-v0.3-Instruct-demi-merge                        | 0.904       | 0.774                | 0.058       | 0.522                |
| Mistral-7B-v0.3-Brain                                      | 0.340       | 0.983                | 0.481       | 0.016                |
| Mistral-7B-v0.3-Instruct-GPTQ-4bit                         | 0.002       | 0.281                | 0.089       | 0.784                |
| Mistral-7B-distilled-from-deepseek-r1-qwen32b              | 0.009       | 0.001                | 0.511       | 0.015                |
| Gemma-2-2b                                                 | 0.747       | 0.869                | 0.689       | 0.000                |
| Gemma-2-2b-it                                              | 0.944       | 0.039                | 0.626       | 0.060                |
| Gemma-2-baku-2b                                            | 0.616       | 0.231                | 0.176       | 0.464                |
| Gemma-2-2b-neogenesis-ita                                  | 0.002       | 0.536                | 0.079       | 0.980                |
| Gemma-2-2b-merged                                          | 0.951       | 0.012                | 0.651       | 0.075                |
| Gemma-2-2b-it-lora-sql                                     | 0.236       | 0.408                | 0.000       | 0.000                |
| Gemma-2-2B-it-4Bit-GPTQ                                    | 0.945       | 0.159                | 0.000       | 0.008                |
| Gemma-2-2b-it-distilled                                    | 0.482       | 0.573                | 0.271       | 0.276                |
| Llama-2-7b                                                 | 0.443       | 0.470                | 0.148       | 0.276                |
| Llama-2-7b-chat                                            | 0.528       | 0.868                | 0.424       | 0.766                |
| tulu-2-7b                                                  | 0.125       | 0.272                | 0.170       | 0.888                |
| EYE-Llama_qa                                               | 0.657       | 0.244                | 0.674       | 0.968                |
| coma-7B-v0.1                                               | 0.953       | 0.004                | 0.573       | 0.731                |
| llama2-Better-Tune                                         | 0.136       | 0.699                | 0.469       | 0.860                |
| Llama-2-7B-Chat-GPTQ                                       | 0.594       | 0.603                | 0.582       | 0.424                |
| llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2 | 0.349       | 0.224                | 0.366       | 0.976                |

Table 15: **Sample results.** Pairwise similarity ( $1 - p$ ) between target models (rows) and base models (columns). Cells with  $1 - p \geq 0.99$  are highlighted in green as same-lineage pairs; the green block-diagonal structure demonstrates that high scores concentrate precisely on the true within-family lineage pairs and nowhere else.